

Global Facilities - Design & Construction Standard - CSA - Admin Office and Amenities Building

ECN Facility:	GLOBAL FACILITIES	ECN Area:	ENGINEERING
Owner(s)	ENGINEERING	Target Audience:	All Facilities
Document Type:	Standard		
Version:	1	Document ID (pdf):	A3YRXSD74VDV-57553043-476
Current ECN:	101113082	Last Modified:	02/09/2022

[ECN History](#)

[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document provides a Required Standard and Guideline on material and design specification that shall be followed and complied in the Design, Engineering and Construction of Admin Office & Amenities Building as part of any green and/or brown field construction projects contract requirement.
- 1.2 The standard is applicable for any green and/or brown field construction projects with regards to provide Admin Office & Amenities Building as per required in the contract.
- 1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.
- 1.4 MasterSpec provided by the America Institute of Architects (AIA) (Refer to [America Institute of Architects \(AIA\) - MasterSpec](#) in **Section 5**) shall form the basis of Micron's Global Design and Construction Specifications Standard.
- 1.5 The established Specifications Standard for each locale shall also comply with Micron's in-house:
 - 1.5.1 RBA Code of Conduct requirements for Semiconductor Manufacturing
 - 1.5.2 Environmental, Health & Safety Standards
 - 1.5.3 Operational Requirements/SOP (Standard Operation Procedures)
 - 1.5.4 Security Protocol

- 1.6 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws, and reflects the current semi-conductor industry's best practices.
- 1.7 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

2.0 Scope

- 2.1 This document for Admin Office & Amenities Building that covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, roof structure, roof finishes, external wall enclosure, internal wall and partition, finishes, Interior Design (ID) works, workstations & cabinets, fixtures and fitting, miscellaneous works to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.
- 2.2 This document applies to all Micron Facilities employees, contractors and vendors at all Micron sites.

3.0 Definitions and Acronyms

3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
AE	External Architectural & Engineering Consultants
DPT	Micron's Designated Project Team
EHS	Micron's Environmental Health & Safety Team
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM's
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFCT	Micron's Global Facilities Construction Team
GFTT	Micron's Global Facilities Technology Team; Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
QP/PE	External Qualified Person (professional architect/engineer)
SME	Micron's Subject Matter Expert
TM	Micron's Team Members

4.0 Admin Office & Amenities Building Design Requirement

- 4.1 All Admin Office & Amenities Building material and design specifications for any green and/or brown field construction project shall comply to local codes & regulations.
- 4.2 The Design Program shall comply with a programming outlined guided by an experiential "Journey Map" within the Micron's Campus for multiple user groups consisting of Micron's Team Member, our Customers & external Visitors. Based on the detailed programming requirements on Micron Workspace Standard (Refer to [Micron's Workplace Standard 2020](#) in **Section 5**), key space requirements are categorized as follows:

4.2.1 "Explore" Space

- 4.2.1.1 Reception Lobby
- 4.2.1.2 Customer Experience Center

- 4.2.1.3 Micron Store
 - 4.2.1.4 HR Kiosk
 - 4.2.1.5 IT2go
 - 4.2.1.6 Product & Innovation room
 - 4.2.1.7 Learning Spaces
 - 4.2.1.8 Patent Wall
- 4.2.2 "Focus" Space
 - 4.2.2.1 Focus Rooms
 - 4.2.2.2 Offices
 - 4.2.2.3 Workstations / Cubicles
 - 4.2.2.4 Labs/Fabs/Benches
 - 4.2.2.5 Co-working spaces
 - 4.2.2.6 Touchdown spaces
- 4.2.3 "Learn" Space
 - 4.2.3.1 Training Rooms
 - 4.2.3.2 Multipurpose Rooms
- 4.2.4 "Collaboration" Space
 - 4.2.4.1 Shared Collaboration Spaces
 - 4.2.4.2 Training Rooms
 - 4.2.4.3 Multipurpose Rooms
- 4.2.5 "Recharge" Space
 - 4.2.5.1 Café
 - 4.2.5.2 Wellness Space
 - 4.2.5.3 Game Room
 - 4.2.5.4 Fitness Room
 - 4.2.5.5 Inter-faith room
 - 4.2.5.6 Mother's Rooms
 - 4.2.5.7 Child Care
 - 4.2.5.8 Outdoor spaces
- 4.3 Depending on the land size and specific operational needs, a reduced program without all the above-mentioned facilities is possible. Nevertheless, the AE shall strive to provide a suitable design process flow that could still reflect Micron's conceptual "Journey Mapping".
- 4.4 The respective floor area of the above-mentioned facility will be site-specific, according to the design brief provided by Micron's stakeholders. The AE shall assist to provide suitable floor area recommendation on facilities that the actual user size cannot be effectively determined by the stakeholders in the design brief, for Micron's DPT review and acceptance at the conceptual design review phase.
- 4.5 Innovative Work Environment
 - 4.5.1 Micron's is striving to nurture a passionate and innovative work culture for the Team Members (TM). The AEs are encouraged to provide a design

approach for the office and amenity spaces to echo a “calm, lively and interactive” ambience. This to help create greater opportunities for spontaneous social meets and cross-discipline collaboration in a relatively stress-free environment to inspire greater creativity and productivity.



Figure 1
*Lush greenery semi- outdoor
office terraces*



Figure 2
Interactive spaces for social meets

4.6 Sustainability Awareness

4.6.1 All Micron facilities should achieve LEED Gold, and push for LEED Platinum where possible, especially for non-manufacturing facilities.

4.6.2 Micron aims high on its environmental impact reduction program in its manufacturing processes. Integrate the sustainability vision in the workplace is one of our objectives. To showcase its commitment, AEs are encouraged to integrate environmental-friendly building features in the campus building architecture to gain visibility and prominent with the public. Some of the plausible features are as follows:

4.6.2.1 Solar and wind power-harvesting system where possible

4.6.2.2 Greenwall supported by rainwater harvesting and auto-irrigation system

4.6.2.3 Lush green semi-outdoor balconies, sky-terraces and alfresco dining space

4.6.2.4 Natural-ventilated or partially mechanical-assist ventilation Reception Lobbies/Atriums

The AE needs to provide actual energy-saving data (such as solar harvesting efficiency index and sun-path simulation) for adopting these sustainability features to validate the proposal and the associated ROI.

4.7 Inclusive Design Standard

4.7.1 Micron is committed to provide an inclusive environment/workspace for all its TM, external vendors, and visitors.

We cannot expose high-hazard activities in our wafer manufacturing and associated facilities to the physically challenged. Notwithstanding, all common amenity and offices spaces outside these production zones should be made barrier-free where reasonably possible.

Micron's Inclusive Design Standard (Refer to [Micron's Global Inclusive Design Standard](#) in **Section 5**) may be over-and-above certain local code requirements as we try to establish a policy to implement a common company standard across all our campuses globally.

4.8 RBA (Responsible Business Alliance) certification

4.8.1 Micron is a member of the RBA for semi-conductors manufacturing, and we conduct period inspection to gain re-certification across all our facilities globally.

The AE shall provide a design for all our campus facilities, including Admin Offices and amenity buildings/structures with this compliance.

4.8.2 The AE shall review the Micron's program brief, and subsequently self-audit all AE design submissions to ensure relevant RBA requirements & guidelines (for semi-conductors manufacturing) are designed & catered for.

4.8.2.1 An example is the RBA Emergency Egress Route Door Guidelines (Refer to [RBA Guideline for Emergency Egress Route](#) in **Section 5**) where panic-bolt push-bar door shall be provided for along all emergency escape route for high-hazard spaces and compartments with ≥ 50 occupancy. This shall include Admin Offices and other amenities spaces within Micron's campus.

Figure 3

Fire-rate door with panic-bolt push-bar and vision glass



5.0 References

- 5.1 [America Institute of Architects \(AIA\) - MasterSpec](#)
- 5.2 [Micron's Workplace Standard 2020](#)
- 5.3 [Micron's Global Inclusive Design Standard](#)
- 5.4 [RBA Guideline for Emergency Egress Route](#)

6.0 Document Control

6.1 Approval

GLOBAL_FAC_GFTT_APPR

6.2 Notification

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

6.3 Final Viewer Access

All Micron Team Members at all Sites

6.4 Retention

Adherence to the requirements specified in
[Global Facilities - Document Control Procedures](#)

6.5 Review

Biennial

7.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	02/09/22